# Who Chooses the Price?

## Acceptance, feasible choice and welfare in guarded Q-learning markets

Mohammadpayam Teabi

Independent researcher

With Apart Research

Research conducted at the AI Collusion Research Sprint, October 2026. Version 2.

Abstract

A pricing agent can achieve perfect proposal acceptance by selecting only prices its controller permits. Does that score indicate useful economic choice? We separate acceptance, feasible price support and economic outcomes using own-profit tabular Q-learners. An initial duopoly study found that a profit-tolerance guard improved consumer surplus but failed a proposal-retention criterion. A matched interface follow-up raised retention to 100% while preserving approximately one allowed price. We now test eight prospectively specified training configurations, varying market calibration, grid density and normalized tolerance, with eight paired seeds and four controller variants per configuration. Four numerical calibrations cover three dimensionless demand shapes; an affine-equivalent pair is a normalization check. Each checkpoint receives equal-length continuations crossing exploration and value updates, with protection active and removed. Guard-aware learners retain every proposal while mean permitted counts range from 1.0 to 4.0 across the selected configurations. Thus singleton choice is conditional, and acceptance alone does not measure permission or welfare. We report all configuration-level economic effects and uncertainty, plus an independently replayed evaluation record. The contribution is a bounded comparative evaluation using established learning and shielding methods. It does not establish intent, broad market safety, semantic handover, or validated human understanding.

## 1 Introduction

Algorithmic pricing raises concerns about harmful high prices, including outcomes learned without communication [1,2]. A guard can improve customer benefit by changing a seller's proposed price. Judging that intervention requires three separate questions: **Was the proposal accepted? What prices were actually available? Who benefited economically?**

Our original 20%-retention gate was an invented practical criterion, not an established field standard. Its failure motivated a direct interface test and then a broader robustness extension. We retain the original negative result. The follow-up asks where choice really collapses, where it remains larger, and how deployment exploration and value updating affect outcomes. The goal is an inspectable economic comparison, not a claim to a new mitigation method.

## 2 Related Work

Calvano et al. [1] study Q-learning pricing, and Eschenbaum et al. [2] examine transfer to unfamiliar rivals and continued adaptation. Shielding [3, Section 5.2] already discusses crediting executed rewards to rejected proposals, those proposals remaining optimal, executed-action updates, preemptive action restriction and persistent shield dependence. Projection learning [4,5] also studies action aliasing and constrained optimization. None of those general mechanisms or repairs is new here.

The sprint starter [6] includes checkpoint branching and replacement. Platform welfare interventions [8] and calibrated-regret auditing [9] are further relevant prior work. Our lagged one-round profit-loss guard is not a calibrated-regret certificate or legal collusion test. The incremental contribution is a matched evaluation of acceptance, permission and economics, with explicit calibration boundaries and ordinary-controller comparisons.

## 3 Methods

### 3.1 Economy, learners and simultaneous timing

Two firms serve 100 potential buyers, with an outside option of utility zero. For quality v, cost c and logit scale mu, z_i = exp((v - p_i)/mu), demand q_i = 100 z_i/(1 + z_A + z_B), profit = (p_i - c)q_i, and CS = 100 mu log(1 + z_A + z_B). Consumer surplus is a model quantity. Economy-specific continuous symmetric Nash and joint-profit benchmarks determine a grid from Nash minus 10% of their price gap to joint optimum plus 10%. All pure finite-grid Nash pairs and joint-profit pairs are enumerated; no continuous theorem is transferred.

Each firm has private Q values and random draws, observes the previous executed-price pair and earns its own profit. Both proposals are sealed before either current action is revealed. Training lasts 3 million rounds, with alpha .15, gamma .95 and exploration exp(-.000004 t). Initialization uses discounted profit against uniform rival prices; lowest-index ties and two random draws per firm/round are fixed. No cooperation reward or scripted retaliation is provided. Q(s,a) moves toward r + gamma max_b Q(s_next,b).

### 3.2 Controllers and the prospectively specified follow-up

The guard allows grid prices within epsilon of the best one-round profit against the previous rival quote and projects other proposals to the nearest allowed price. Proposal credit considers all proposals; aware learning restricts choice, exploration and next-state max-Q to the feasible action image. A cap clips at the grid price nearest continuous Nash. The new training matrix has no cap-aware arm; a retention comparison against cap-proposal would therefore confound interfaces. Earlier deployment includes cap-aware as a control.

Four calibrations are base (v=2,c=1,mu=.25), mu=.5, v=1.5, and c=.6, changing one parameter at a time. Two share (v-c)/mu=2: four calibrations cover **three demand shapes**, not four independent regimes. Epsilon is .1 times own continuous Nash profit divided by base continuous Nash profit, times .5/1/2; the base factor-1 value is exactly .1.

Eight learned cells comprise all four calibrations at 15 prices/factor 1, plus base grids 11/31 and base tolerance factors .5/2. Each cell has eight paired seeds and four training arms, totaling 256 labels/768 million rounds. Every trained checkpoint is forked to protection active/removed, updates off/on and exploration 0/.01, all for 20,000 rounds (2,048 branches). Full-window outcomes are primary; final 5,000 are dependent diagnostics. Ten-thousand-resample intervals use whole paired seeds, n=8 per cell. No cross-economy raw-CS pooling or family-wise significance claim is made.

# 4 Results

## 4.1 Acceptance and economic choice answer different questions

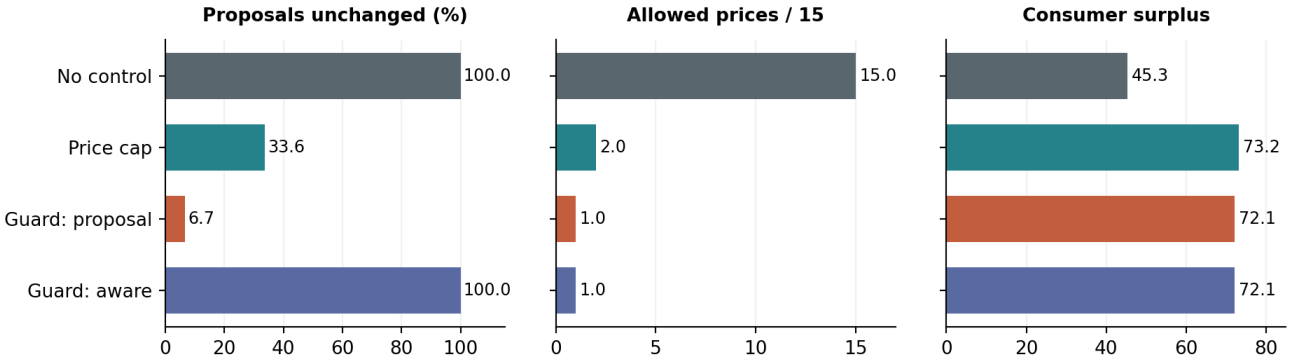



**Figure 1.** Original matched deployment from 12 unconstrained-trained checkpoints, 20,000 adaptive rounds. Aware selection accepts every proposal while allowing about one price. Retention is interface bookkeeping, count is modeled permission, and CS is an economic outcome. None alone establishes useful autonomy.

In the original study, the proposal-credit guard increases CS from 45.2917 to 72.1143. The paired increase 26.8226 has interval [25.3820, 28.3111], exceeding the prespecified 1.940098 welfare threshold (5% of the continuous Nash-to-joint CS gap), but retention 6.7452% fails its 20% threshold. Changing to actual aware selection gives 100% retention and CS 72.1177; the CS difference is only .003431 [.000012, .008063]. The numerical gate is not retuned, and high prices are not treated as proof of intentional collusion.

The original base grid/tolerance permits one price near the terminal state: the second-best grid profit loss .110063 exceeds epsilon .1. Projection idempotence also lets a recorded path be reexpressed with accepted actions as proposals. That preserves its economic path, **not** proposal-index Q updates or a counterfactual trained policy. The actual learned-aware comparison is the behavioral test.

## 4.2 Why one allowed price is a conditional result

For a previous rival price j, sort candidate profit losses as L_(1) <= L_(2) <= ... . The allowed set has exactly one action when L_(1) <= epsilon < L_(2), with the declared numerical tie tolerance. On the base 31-price grid, the next loss near grid Nash is about .012635, below .1, permitting four prices. The changed-shape 15-price examples also permit multiple prices. These are finite-grid facts, not a general autonomy theorem.

A complete geometry sweep covers four calibrations, three grids and three tolerance factors. It enumerates possible prior rival prices, separately from the states learners actually visit (Appendix B). The normalized equivalent calibration pair serves as a units check; finite-precision learned path agreement is measured separately, not assumed.

### 4.3 Equal-budget training across all eight selected configurations

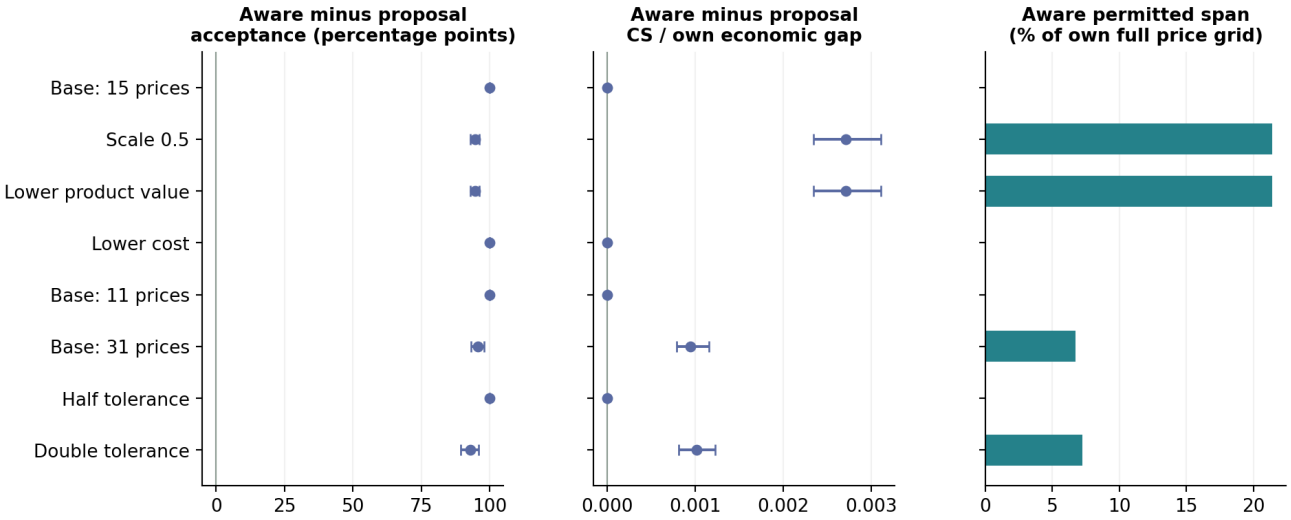



**Figure 2.** New trained arms: eight paired seeds per cell, protection active, updates on, exploration .01, full 20,000 rounds. Points/intervals compare aware and proposal guard interfaces. CS differences are divided by each economy's own Nash-to-joint CS gap; raw levels are not pooled. Bars show allowed span relative to that cell's full grid. The mu=.5 and lower-value rows have equal dimensionless demand shapes.

**Table 1.** Guard-aware permission, customer and seller outcomes. Last column: paired aware-minus-proposal CS [95% interval].

| Configuration | Allowed count | One-price visits | Aware CS | Profit A / B | CS difference [interval] |
|---|---|---|---|---|---|
| Base, 15 prices | 1.00 | 100.0% | 72.119 | 22.02 / 22.02 | 0.000 [0.000, 0.000] |
| Scale 0.5 | 3.99 | 0.0% | 64.748 | 31.19 / 31.19 | 0.070 [0.061, 0.081] |
| Lower product value | 3.99 | 0.0% | 32.374 | 15.59 / 15.59 | 0.035 [0.030, 0.040] |
| Lower cost | 1.00 | 100.0% | 109.348 | 23.84 / 23.84 | 0.000 [0.000, 0.000] |
| Base, 11 prices | 1.00 | 100.0% | 70.658 | 22.67 / 22.67 | 0.000 [0.000, 0.000] |
| Base, 31 prices | 3.02 | 0.0% | 69.004 | 23.40 / 23.40 | 0.037 [0.031, 0.045] |
| Half tolerance | 1.00 | 100.0% | 72.119 | 22.02 / 22.02 | 0.000 [0.000, 0.000] |
| Double tolerance | 2.01 | 0.0% | 68.527 | 23.61 / 23.60 | 0.039 [0.031, 0.048] |

Aware mean permitted counts span 1.00-3.99 across the selected cells. All aware proposals are retained by construction, including exploration, even when allowed support differs. A 100% score therefore cannot distinguish a singleton price from several available prices. Feasible range remains modeled permission, not demonstrated economically valuable choice; profits and realized behavior are retained in the full tables.

These arms train under their own controllers from matched seeds; they do not share final Q tables. Interface support, exploration, bootstrap support and training trajectory change together. The comparison establishes configuration-level behavior, not a sole-cause credit assignment explanation. All assigned cells, seeds and windows are retained.

The cap gives higher primary CS in all eight cells, but this is not dominance on every objective. On the 31-price grid, cap CS is 72.3649 versus aware-guard 69.0044, while average profit per seller is about 21.910 versus 23.398. Similar allowed counts can therefore conceal different permitted price locations and customer/seller tradeoffs.

## 4.4 Separate deployment exploration from updating and enforcement

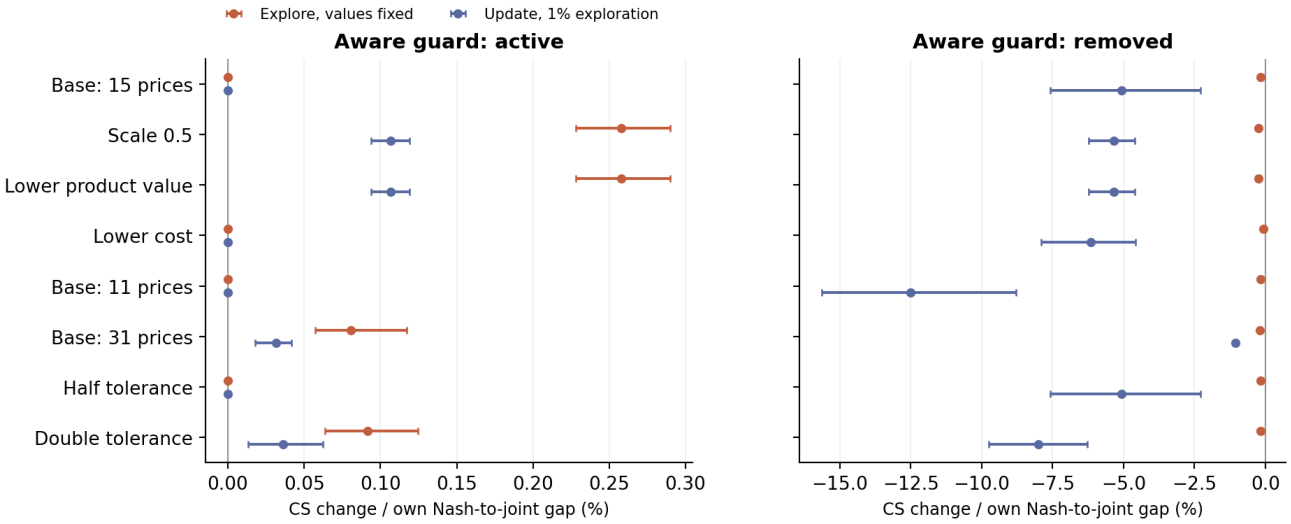



**Figure 3.** New trained aware checkpoints: exploration effects hold Q values fixed; update effects hold exploration at .01. Points/95% paired intervals use the same 20,000-round horizon and initial checkpoint. Changes are normalized by each economy's own Nash-to-joint CS gap; panel scales differ. The complete factorial results also include updates without exploration, all other arms, interactions and final-5,000 windows.

With protection active, adding 1% exploration while freezing Q changes aware-guard CS by cell means from 0.0000 to 0.0668; updating without exploration leaves CS unchanged in all eight cells. On the 31-price grid, exploration alone adds .031351 [.022249, .045435] CS, and updates with exploration add a further .012173 [.006905, .016173]. These are effects in newly controller-trained cohorts, not an explanation of the older cap transient.

**Table 2.** Base aware-guard mean CS in equal-length tests. Each start uses the same trained checkpoint; protection removal changes the action space.

| Behavior during test | Protected CS | Released CS |
|---|---|---|
| Repeat learned choices | 72.119 | 61.308 |
| Explore 1%, no updates | 72.119 | 61.239 |
| Update, no exploration | 72.119 | 52.370 |
| Update and explore 1% | 72.119 | 59.273 |

## Release and historical continuation have distinct scopes

Release removes both mask and projector from the identical trained checkpoint. It changes the effective action space, including previously untrained choices; effects do not establish a transferable unconstrained policy or impossibility of learning one.

The earlier original-checkpoint continuation remains informative: its cap CS lead over the aware guard falls from about 1.07 initially to .00261 in the final 20,000 of 100,000 rounds. The present factorial design starts from newly controller-trained policies. It separates exploration and updating there; it does not identify the cause of that historical transient. Finite-window stability is not proof of convergence.

## 5 Discussion and Limitations

The robust finding is the separation of measurements, not universal collapse to one action. Perfect acceptance can accompany different allowed price sets and economic outcomes. The base singleton example is sharp, but a denser grid and changed demand shape break its specific premise. A useful evaluation should report the proposal interface, allowed count/range, customer and seller outcomes, and protected versus released behavior together. This is an evaluation checklist motivated by our own initial gate, not evidence that the field universally uses a misleading autonomy measure.

### Limits and future work

All models are symmetric two-firm logit economies with private tabular learners, fixed hyperparameters, optimistic initialization and finite training. Four numerical calibrations contain three dimensionless shapes; other parameters are not independently randomized. Eight seed units per cell give limited uncertainty information; many intervals are descriptive, without multiplicity-adjusted superiority claims. Marginal and rival-index conditional entropy describe realized variation, not full-state strategic freedom. High prices do not establish intent, credible punishment or legal collusion. Future tests should use genuinely different demand families, asymmetric costs and richer learners.

New numerical replay checks every saved evaluation transition and aggregate using a separate implementation. It does not independently repeat all 768 million training updates or establish external validity. Earlier continuous-price guarantees remain separate. Semantic handover and language understanding have not been tested.

### Graphical explanation and reproduction

The human game now offers four labeled scenarios, optional first-three-round guidance, editable strategy suggestions, correction previews, readable mobile shop cards and a 30-round debrief. Profit and customer benefit are separate objectives. It preserves the same sealed simultaneous economic rules. Its continuous prices and a frozen saved opponent are distinct from these finite-grid training experiments. Developer desktop, 320/390-pixel, keyboard, replay and completion checks passed; no independent human comprehension or accessibility study was performed.

## 6 Conclusion

Accepted proposals, permitted choice and economic benefit must be assessed separately. The original one-price counterexample survives in its calibration, while selected changes permit several prices under the same 100%-acceptance interface. Matched exploration/update controls and release tests make those conditional outcomes inspectable. The work strengthens a reproducible bounded comparison, with established learning and shielding methods; it does not claim a new broadly effective anti-collusion defense.

## Code and Data

The accompanying evidence archive contains frozen protocols, sources, every seed, lossless checkpoints, four-byte-per-round evaluation traces, full results and independent auditors. Run `node research/reproduce.js` for offline tests and replay; `--quick` skips replay. Regenerate figures separately with `python research/build-execution-figures.py` using the listed free Python packages. On Windows, VERIFY-RESEARCH.cmd runs the full check. The game is started with `node research/demo-server.js`. No public submission is claimed.

## References

[1] E. Calvano, G. Calzolari, V. Denicolo and S. Pastorello. Artificial Intelligence, Algorithmic Pricing, and Collusion. American Economic Review 110(10), 3267-3297, 2020. https://www.aeaweb.org/articles?id=10.1257/aer.20190623

[2] N. Eschenbaum, F. Mellgren and P. Zahn. Robust Algorithmic Collusion. arXiv:2201.00345, 2022. https://arxiv.org/abs/2201.00345

[3] M. Alshiekh et al. Safe Reinforcement Learning via Shielding. AAAI 2018; arXiv:1708.08611. https://arxiv.org/abs/1708.08611

[4] H. Markgraf, S. Sawant, H. Krasowski, L. Schafer, S. Gros and M. Althoff. Safe Reinforcement Learning using Action Projection: Safeguard the Policy or the Environment? arXiv:2509.12833, 2025, revised 2026. https://arxiv.org/abs/2509.12833

[5] S. Gros, M. Zanon and A. Bemporad. Safe Reinforcement Learning via Projection on a Safe Set: How to Achieve Optimality? IFAC-PapersOnLine 53(2), 8076-8081, 2020. https://doi.org/10.1016/j.ifacol.2020.12.2276

[6] Poseidon Research. AI Collusion Research Sprint starter repository. https://github.com/poseidon-research/collusion_hackathon

[7] Apart Research. AI Collusion Research Sprint, October 23-25, 2026. Event guidelines and public report template. https://apartresearch.com/sprints/ai-collusion-research-sprint-2026-10-23-to-2026-10-25

[8] G. Brero, E. Mibuari, N. Lepore and D. C. Parkes. Learning to Mitigate AI Collusion on Economic Platforms. Advances in Neural Information Processing Systems 35, 2022. https://papers.neurips.cc/paper_files/paper/2022/file/f746974abd33c0015ca583a267dac1fd-Paper-Conference.pdf

[9] J. D. Hartline, C. Wang and C. Zhang. Regulation of Algorithmic Collusion, Refined: Testing Pessimistic Calibrated Regret. arXiv:2501.09740, 2025. https://arxiv.org/abs/2501.09740

## Author Contributions

Mohammadpayam Teabi directed the project, supplied its original game context and authorized the local work. AI development agents assisted formulation, implementation, execution, analysis, literature checking, adversarial review and drafting. Separate AI-agent review is not human peer review. Established economics, shielding and Q-learning are attributed above. The paper follows the accessible public template's structure and adds the required ethical appendix [7].

# Appendix A. Limitations and Dual-Use / Ethical Considerations

## A.1 Evidence and independent units

The initial protocol and amendment, subsequent credit/interface study and this robustness extension are distinct chronological stages. The original outcomes motivated follow-ups; later source-hash freezes do not constitute independent preregistration of earlier claims. Local manifest times establish ordering, not third-party registration. Eight seeds 7001-7008 are paired across variants and reused across configuration cells. There are 256 training labels, not 256 independent comparative replications. Full and last-5,000 windows share their parent paths; cross-cell effects are dependent. No failed cells were silently dropped.

The new study uses 768 million training rounds and 40.96 million saved evaluation transitions over 2,048 branches. Checkpoint forks hold Q tables, prior prices, clocks and PRNG states fixed at each within-arm start. A separate auditor reconstructs economic formulas, benchmark/guard geometry, selections, random draws, evaluation Q updates, bins, final checkpoints and aggregates; injected corruptions test rejection. Full training trajectories are not independently replayed. Version-1 evidence and its closed report remain preserved.

## A.2 Scope, assumptions and possible harm

The experiment is a stylized symmetric deterministic duopoly. Known demand/cost parameters give the controller privileged model knowledge; uncertainty, strategic misreporting, asymmetry, richer action spaces and real customer behavior remain outside scope. The implemented lagged constraint does not limit actual regret against an unseen simultaneous rival price. Numerical replay and a playable explanation do not establish legal compliance, real-world safety or human understanding.

Pricing experiments could aid anticompetitive agent design or teach ways to make compliance metrics look favorable. We report controller restrictions, economic costs and ordinary baselines together with negative outcomes. The package contains a toy environment and no customer data or deployment certification. Earlier exploratory account-backed pilots consumed 342 requests and are excluded from these results. Current experimental sellers use local tabular learning, with no LLM/API inference, purchased compute or paid service.

## A.3 Pre-existing work and AI assistance

Earlier bank/auction games predate the sprint. The guarded-pricing tabular studies, current robustness extension, audit tooling and report were developed during it, adapting known market and learning ingredients. Pre-existing material is identified in repository history and attribution notes. The report does not treat earlier semantic-handover scripts as evidence of language understanding.

# LLM Usage Statement

Codex research and coding agents assisted conceptualization, implementation, local runs, analysis, primary-source review, adversarial feedback and drafting. This text is substantially AI-drafted under the author's direction; it is not represented as primarily human-written. Independent reviews are separate AI-agent checks, not unperformed human replication. Claims were checked against saved data and sources, with independently implemented evaluation replay. The experimental sellers themselves made no model/API calls. The author remains responsible for the report and submission.

# Appendix B. Reading guard geometry and normalization

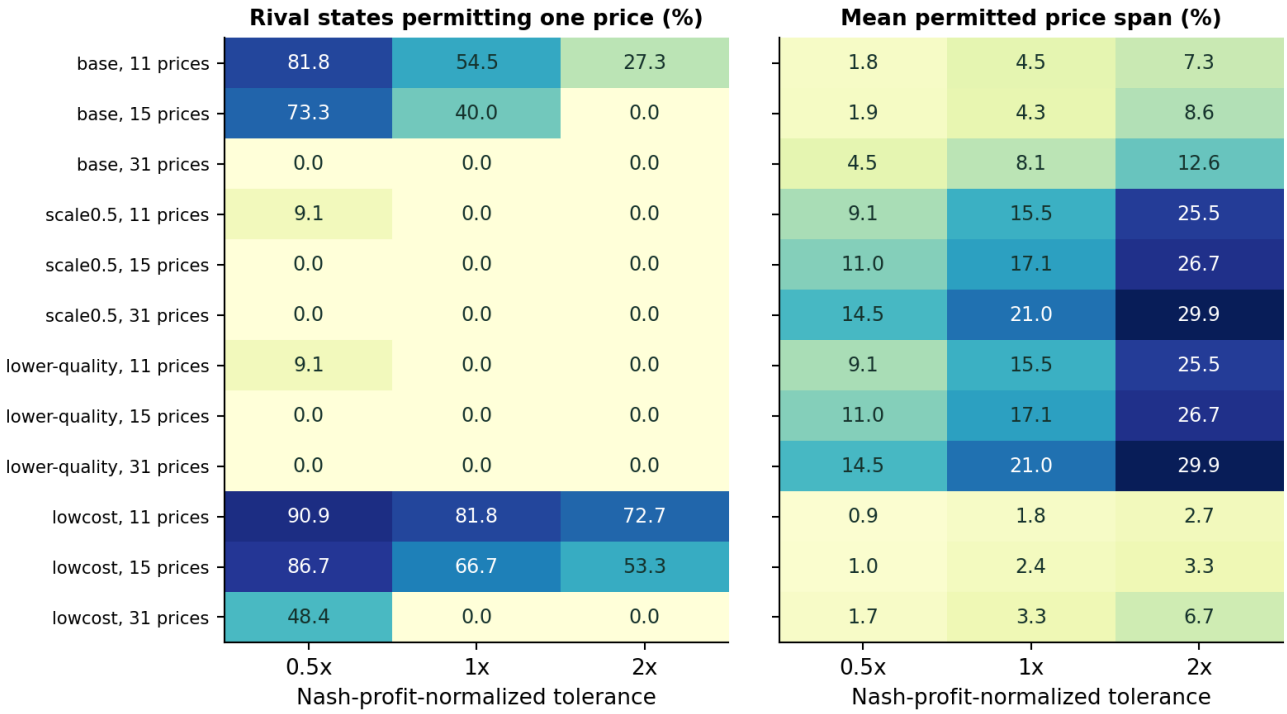



**Figure B1.** All 36 geometry cells. Each prior rival grid price receives equal weight; these are off-path possibilities, not observations from 36 trained experiments. Singleton frequency and allowed span answer different questions. Span is divided by each grid's full price range. An agent's visited-state distribution can emphasize different parts of this geometry, so Figure 2 reports on-path measures separately.

Write x = (p-c)/mu and theta = (v-c-outside)/mu. Demand depends on theta and x; profits and CS scale by 100 mu. Equal-theta calibrations with equal normalized tolerance therefore have equivalent normalized geometry in ideal arithmetic. Here the mu=.5 and lower-quality calibrations both have theta=2, compared with base theta=4 and lower cost theta=5.6. This explains why four calibrations cover three shapes. Finite-precision Q-learning trajectories can differ and are not counted as independent economic diversity. The separately labeled post-freeze invariance diagnostic records actual agreement and differences.

For each reported comparison, read the full seed table, both seller profits, allowed count/span, singleton-visit frequency, and execution entropy together. Bootstrap intervals measure seed variation within these fixed configurations. Zero entropy does not by itself prove a controller removed choice; an unconstrained deterministic policy may also select one action. Aware retention is structurally 100% and must not be interpreted as an algorithmic performance win.